

wifi_air_suite
PRODUCT GUIDE - v3.0.0

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Contents

1. Overview	2
2. Installation and package layout	3
3. Execution model and safety gates	3
4. Capture fundamentals	3
5. Targeted BSSID capture with <code>--channel</code>	4
6. Interval capture and long-running workflows	4
7. Post-processing and generated artifacts	5
8. Exclusions and OUI manufacturer enrichment	5
9. Logging modes and <code>--nolog</code>	6
10. Inspection and explicit cracking workflows	6
11. Authorized-lab actions	6
12. Runtime layout and storage management	7
13. Complete option reference	7
14. Worked examples	8
15. Troubleshooting	8
16. Frequently asked questions	9
17. Version history and release summary	9

`wifi_air_suite` is a unified Wi-Fi capture, inspection, post-processing and authorized-lab toolkit for Linux/Kali. The v3.0.0 release is a clean major baseline built from the validated v2.1.4 implementation. It preserves the established operational behavior while bringing the script, examples, specifications, installation notes, rationale and product documentation onto one release baseline.

Active wireless actions are intended only for networks and devices you own or are explicitly authorized to test.

1. Overview

1.1 What v3.0.0 is

v3.0.0 is the major-release baseline of the current `wifi_air_suite` CLI. The major version does not hide a behavior rewrite: it promotes the validated v2.1.4 implementation into a fully consolidated release with current documentation and a complete Product Guide.

1.2 Core principles

- Capture stays capture-only: `--capture` runs `airodump-ng` and never implicitly starts cracking or active actions.
- Real operational work requires the explicit `--exec` gate.
- `--simulate` provides a non-operational dry run.
- Exactly one action is selected per invocation.
- Long captures can be sliced with `--interval`.
- Persistent terminal logs can be disabled with `--nolog` while CAP/CSV and post-processing outputs remain available.
- Manufacturer data can be refreshed with `--update-oui`.
- A known BSSID can be pinned to a known channel with `--channel N`.

1.3 Main use cases

Use case	Relevant features
General RF inventory	<code>--capture</code> , multi-channel hopping
Focused AP observation	<code>--bssid</code> , optional <code>--channel</code>
Long investigation	<code>--interval</code> , <code>--post-process</code> , <code>--nolog</code>
Manufacturer attribution	<code>--update-oui</code> , enriched CSV
Noise reduction	<code>myinfo/exclusions.txt</code>
Capture inspection	<code>--check</code>
Explicit wordlist test	<code>--crack</code> , <code>--wordlist</code> , <code>--bssid</code>
Authorized lab validation	<code>--attack-hosts</code> , <code>--attack-preauth</code> , <code>--test-injection</code>

2. Installation and package layout

2.1 Files in the complete package

```
wifi_air_suite.sh
set_unset_to_monitor.sh
README.md
INSTALL.md
SPECIFICATIONS.md
EXAMPLES.md
WHY.md
CHANGELOG.md
PRODUCT_GUIDE.md
wifi_air_suite_v3.0.0_Product_Guide.pdf
.gitignore
myinfo/
  exclusions.txt
  exclusions_enrichi.txt
  exclusions_oui_resolved.txt
```

myinfo/oui.txt is intentionally not shipped as a placeholder. Populate it from the maintained IEEE source with `./wifi_air_suite.sh --update-oui`.

2.2 Permissions and first checks

```
chmod +x wifi_air_suite.sh set_unset_to_monitor.sh
./wifi_air_suite.sh --version
./wifi_air_suite.sh --prerequis
./wifi_air_suite.sh --init
./wifi_air_suite.sh --update-oui
./wifi_air_suite.sh --doctor
```

2.3 Expected platform

The suite targets Linux/Kali and expects the Aircrack-ng ecosystem plus standard networking and GNU utilities. The script can install its supported package set through `--install`.

3. Execution model and safety gates

3.1 Real execution

```
./wifi_air_suite.sh --exec --capture [options]
```

Operational actions require `--exec`.

3.2 Simulation

```
./wifi_air_suite.sh --simulate --capture [options]
```

Simulation resolves and prints the requested configuration without performing privileged/system/network changes.

3.3 One action per invocation

The parser rejects conflicting actions. `--capture`, `--check`, `--crack`, `--attack-hosts`, `--attack-preauth` and `--test-injection` remain explicit and separate.

4. Capture fundamentals

4.1 Five-minute capture

```
./wifi_air_suite.sh --exec --capture \  
  --interface wlan0 \  
  --duration 300
```

4.2 Infinite capture

```
./wifi_air_suite.sh --exec --capture \  
  --interface wlan0 \  
  --infinite
```

4.3 Default multi-channel behavior

Without `--channel`, the capture engine uses this established channel set:

```
1,2,3,4,5,6,7,8,9,10,11,12,13,14,  
36,40,44,48,52,56,60,64,  
100,104,108,112,116,120,124,128,132,136,140,144,  
149,153,157,161,165
```

The command also enables manufacturer display, ACK display, beacons, uptime, WPS information, CSV/PCAP output and periodic write updates.

5. Targeted BSSID capture with `--channel`

5.1 Why it exists

When both the BSSID and its RF channel are already known, hopping through unrelated channels wastes dwell time. v3.0.0 therefore includes the targeted-channel option introduced in the validated v2.1.4 baseline.

5.2 Exact example

```
./wifi_air_suite.sh --exec --capture \  
  --interface wlan0 \  
  --bssid C6:4F:D5:24:D3:1E \  
  --channel 4 \  
  --duration 300 \  
  --post-process \  
  --nolog
```

With `--channel 4`, airodump-ng is given only channel 4. Without `--channel`, the normal multi-channel list is preserved.

5.3 Validation rules

`--channel`:

- is valid only with `--capture`;
- requires `--bssid MAC`;
- accepts only channels already supported by the capture engine;
- does not change the behavior of unrelated actions.

6. Interval capture and long-running workflows

`--interval N` is in minutes; `--duration` remains in seconds.

6.1 One hour in 10-minute slices

```
./wifi_air_suite.sh --exec --capture \  
  --interface wlan0 \  
  --duration 3600 \  
  --interval 10 \  
  --post-process \  
  --nolog
```

This produces six independent slices. Every slice receives a new collision-safe prefix and normal `-01.cap` / `-01.csv` files.

6.2 Infinite sliced capture

```
./wifi_air_suite.sh --exec --capture \  
  --interface wlan0 \  
  --infinite \  
  --interval 10 \  
  --post-process \  
  --accept \  
  --nolog
```

Monitor mode is entered once and retained between slices. Stop with Ctrl+C or with `--stop` from another terminal when the runtime PID is registered.

7. Post-processing and generated artifacts

With `--post-process`, each completed slice can generate:

<code>.results/csv/</code>	copied raw CSV
<code>.results/filtered/</code>	filtered CSV + filtered Markdown
<code>.results/enriched/</code>	manufacturer-enriched CSV
<code>.results/generated/</code>	generated/copy artifacts

7.1 Filtered Markdown

The Markdown is generated from the already filtered CSV, so the CSV and Markdown represent the same filtered data set. Access points and stations are rendered as separate Markdown tables.

7.2 Open automatically in Kate

```
./wifi_air_suite.sh --exec --capture \  
  --interface wlan0 \  
  --duration 300 \  
  --post-process \  
  --open-kate
```

`--open-kate` requires `--post-process` and launches Kate asynchronously.

8. Exclusions and OUI manufacturer enrichment

8.1 Exclusions

The default exclusion source is:

`myinfo/exclusions.txt`

MAC addresses from this file are removed from filtered/post-processed results. Raw RF capture remains untouched.

The complete package also carries the human-readable companion files:

`myinfo/exclusions_enrichi.txt`
`myinfo/exclusions_oui_resolved.txt`

8.2 Override exclusions

```
./wifi_air_suite.sh --exec --capture \  
  --interface wlan0 \  
  --duration 300 \  
  --post-process \  
  --exclusions-file /path/to/custom_exclusions.txt
```

8.3 Update OUI manufacturer data

```
./wifi_air_suite.sh --update-oui
```

The updater downloads the IEEE Registration Authority MA-L/OUI listing, normalizes it to the format used by the suite, validates that at least 10,000 entries were recognized, backs up the previous local file and installs the new database only after validation succeeds.

Simulation:

```
./wifi_air_suite.sh --simulate --update-oui
```

9. Logging modes and --nolog

9.1 Default logging

Normal operation can create a global log and an airodump terminal log under `.results/logs/`. Live non-spinner capture uses a PTY so the full airodump display remains visible while being recorded.

9.2 No persistent logs

```
./wifi_air_suite.sh --exec --capture \  
--interface wlan0 \  
--duration 3600 \  
--post-process \  
--nolog
```

--nolog / --no-log disables persistent `.log` files for that invocation. It does **not** disable CAP, CSV, filtered CSV, filtered Markdown, enriched CSV or generated outputs.

10. Inspection and explicit cracking workflows

10.1 Check a capture

```
./wifi_air_suite.sh --exec --check --offset 1
```

Or target one capture and BSSID:

```
./wifi_air_suite.sh --exec --check \  
--cap-file .results/cap/capture.cap \  
--bssid AA:BB:CC:DD:EE:FF
```

The check is bounded and non-interactive.

10.2 Explicit wordlist action

```
./wifi_air_suite.sh --exec --crack \  
--cap-file .results/cap/capture.cap \  
--bssid AA:BB:CC:DD:EE:FF \  
--wordlist /path/to/wordlist.txt
```

--crack requires both a BSSID and a wordlist and is never started implicitly by capture.

11. Authorized-lab actions

The suite keeps active wireless operations behind explicit actions. They are intended for owned/authorized lab environments.

11.1 Deauthentication from selected CSV pairs

```
./wifi_air_suite.sh --simulate --attack-hosts \  
--interface wlan1 \  
--csv-file .results/cap/lab-01.csv \  
--bssid AA:BB:CC:DD:EE:FF \  
--station 11:22:33:44:55:66
```

Use simulation to verify selection before any authorized real execution.

11.2 Fake authentication and injection test

The distinct actions are:

```
--attack-preauth  
--test-injection
```

They never run as side effects of --capture.

12. Runtime layout and storage management

Default runtime root:

```
.results/  
  cap/  
  csv/  
  filtered/  
  enriched/  
  generated/  
  logs/  
  tmp/  
  archive/
```

12.1 Custom result disk

```
./wifi_air_suite.sh --exec --capture \  
  --interface wlan0 \  
  --infinite --interval 10 --post-process --nolog \  
  --dest-dir /mnt/data3_81g/wifi_results
```

12.2 Maintenance

```
./wifi_air_suite.sh --list-captures  
./wifi_air_suite.sh --list-csv  
./wifi_air_suite.sh --clean-tmp  
./wifi_air_suite.sh --purge  
./wifi_air_suite.sh --stop
```

--purge manages runtime artifacts only; myinfo/ and source/documentation files are preserved.

13. Complete option reference

Option	Purpose
--help, -h	Complete help
--version	Current version
--changelog, -ch	Internal changelog
--exec, -exe	Real operational gate
--simulate, -s, --dry-run	Non-operational simulation
--prerequis, -pr	Prerequisite check
--install, -i	Install supported packages
--doctor	Diagnostics
--init	Initialize layout
--show-paths	Show resolved paths
--print-config	Show resolved configuration
--update-oui	Refresh manufacturer database
--capture	Capture-only action
--check	Bounded inspection
--crack	Explicit wordlist test
--attack-hosts	Authorized/lab deauth action
--attack-preauth	Authorized/lab fakeauth action
--test-injection	Authorized/lab injection test
--interface IFACE	Wi-Fi interface
-d, --duration SECONDS	Finite capture duration
--infinite	Unlimited capture
--interval MINUTES	Independent capture slices
-b, --bssid MAC	BSSID selection/filter
--channel N	Single channel for targeted BSSID capture
--station MAC	Station/client selection
--post-process	Generate analytical outputs
--no-post-process	Disable post-processing
--open-kate	Open filtered Markdown
--accept	Compatibility acceptance mode

Option	Purpose
--archive-old	Archive older runtime outputs
--spinner	Lightweight timed-capture display
--no-spinner	Full live display
--nolog, --no-log	Disable persistent logs
-o, --offset N	Select Nth newest file
--all	Select all CSV files where supported
--cap-file FILE	Explicit capture file
--csv-file FILE	Explicit CSV file
-X, --exclusions-file FILE	Override exclusions
-w, --wordlist FILE	Wordlist for explicit crack
--deauth-count N	Authorized deauth frame count
--dest_dir, --dest-dir DIR	Override runtime root
--stop, -st	Stop registered invocation
--clean-tmp	Clean temporary runtime files
--purge, -pu, --clean-runtime	Purge managed runtime files

14. Worked examples

A. Fast five-minute inventory

```
./wifi_air_suite.sh --exec --capture --interface wlan0 \
  --duration 300 --post-process --nolog
```

B. Focused five-minute BSSID/channel capture

```
./wifi_air_suite.sh --exec --capture --interface wlan0 \
  --bssid C6:4F:D5:24:D3:1E --channel 4 \
  --duration 300 --post-process --nolog
```

C. One-hour sliced investigation

```
./wifi_air_suite.sh --exec --capture --interface wlan0 \
  --duration 3600 --interval 10 \
  --post-process --accept --nolog
```

D. Infinite overnight workflow

```
./wifi_air_suite.sh --exec --capture --interface wlan0 \
  --infinite --interval 15 \
  --post-process --accept --nolog
```

E. Refresh OUI before analysis

```
./wifi_air_suite.sh --update-oui
./wifi_air_suite.sh --exec --capture --interface wlan0 \
  --duration 600 --post-process --nolog
```

15. Troubleshooting

Symptom	Likely cause	Action
Interface not found	Wrong interface name or monitor transition issue	<code>./wifi_air_suite.sh --doctor;</code> <code>inspect iw dev</code>
--channel rejected	Missing BSSID, wrong action or unsupported channel	Use with <code>--capture --bssid;</code> verify supported list
No filtered Markdown	<code>--post-process</code> not enabled	Add <code>--post-process</code>
Kate does not open	Kate missing or <code>--open-kate</code> used without post-process	Install Kate; use both options
Manufacturer is Unknown	OUI file missing/stale	Run <code>--update-oui</code>
Logs consume too much space	Persistent PTY logging enabled	Add <code>--nolog</code> for future captures

Symptom	Likely cause	Action
No CAP selected by check	Runtime root/offset mismatch	Use <code>--list-captures</code> , <code>--cap-file</code> , or correct <code>--dest-dir</code>
Sudo prompt/action fails	No usable controlling terminal or ticket	Run from a normal terminal; use <code>--doctor</code>
Post-process output still contains known MAC	Exclusion file incomplete or wrong override	Inspect <code>myinfo/exclusions.txt</code> and <code>-X</code> path

16. Frequently asked questions

Does `--channel` replace `--bssid`?

No. It complements it. The current contract intentionally requires `--bssid` whenever `--channel` is supplied.

What happens if I omit `--channel`?

Nothing changes: the established multi-channel list is used.

Does `--nolog` mean no results?

No. It disables persistent `.log` files only. CAP/CSV and analytical post-processing artifacts remain available.

Does capture automatically crack?

No. Capture, check and crack are separate actions.

Does `--post-process` change raw capture data?

No. Filtering and enrichment operate on copied/derived CSV artifacts. Raw CAP/CSV capture remains available.

Why is `oui.txt` not shipped in the release package?

To avoid replacing a user's maintained local database with a stale or placeholder file. Use `--update-oui` to create/update the local database from the IEEE source.

Where are all command examples?

`EXAMPLES.md` is the exhaustive command catalog and should be treated as the detailed CLI cookbook.

17. Version history and release summary

Version	Summary
v3.0.0	Major consolidated baseline; complete documentation/Product Guide; preserves validated v2.1.4 behavior
v2.1.4	Added targeted BSSID <code>--channel N</code> capture
v2.1.3	Added validated <code>--update-oui</code> workflow
v2.1.2	Added <code>--nolog</code> / <code>--no-log</code>
v2.1.1	Monitor helper resolved beside main script
v2.1.0	Filtered Markdown and optional Kate opening
v2.0.0	Interval capture and per-slice post-processing

v3.0.0 release checklist

- Main script reports v3.0.0.
- Bash syntax validates successfully.
- `--channel` parser/validation/build path is present.
- Existing multi-channel behavior remains the default.
- README, INSTALL, SPECIFICATIONS, EXAMPLES, WHY and CHANGELOG identify v3.0.0 as the current release.
- Product Guide is included in Markdown and PDF form.

- The three validated exclusion files are included unchanged from the supplied v2.1.4 package baseline.
- No checksum file is included.
- No placeholder `oui.txt` is included.

For exact CLI edge cases and the largest command catalog, see `EXAMPLES.md`. For design contracts, see `SPECIFICATIONS.md`. For installation, see `INSTALL.md`. For release history, see `CHANGELOG.md`.